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Allocating camera views and time instants for trajectory-length measurement under a frame-processing budget

Bowen Tan¹, Zihan Li³, Shengjie Guo⁴, Yi Sui², Xiaolei Zhang², Yi Li² and Jun Ji^{2,*}¹The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong SAR, China²College of Computer Science and Technology, Qingdao University, 308 Ningxia Road, Qingdao 266071, China³College of Mathematics and Statistics, Qingdao University, 308 Ningxia Road, Qingdao 266071, China⁴Inner Mongolia Agricultural University, 306 Zhaowuda Road, Saihan District, Hohhot 010018, Inner Mongolia, China

*Author to whom any correspondence should be addressed.

E-mail: junji@qdu.edu.cn

Keywords: vision-based measurement, distance travelled, trajectory length, multi-camera triangulation, sampling design, measurement uncertainty, budget allocation**Abstract**

A multi-camera trajectory-length measurement can divide a finite processing budget between simultaneous views and sampling instants. We evaluate this allocation using released image centroids from a seven-camera rig and an independent optical tracking reference. Original protocol-frozen results are distinguished from exploratory analyses on four previously examined records. A calibration-selected three-view configuration at 28 instants gives a mean absolute error of 9.60 mm, compared with 50.77 mm for seven views at 12 instants, at exactly 84 processed frames per window. This ordering persists under three observation-noise settings, reference decimation and deletion of each record in turn. However, a broader grid of budget caps and two to seven views reveals a reversal: at 168 frames, the tested three-view schedule has 85.66 mm error versus 10.55 mm for seven views, driven by rare extreme errors. Thus spending an increased budget on more instants can worsen performance, and no universal three-view prescription follows. Comparisons report actual frame use, completed-window error, completion and tail risk. Overlapping-camera noise controls constrain possible explanations without identifying a causal saturation mechanism. A tested curvature-based sampling rule is evaluated on explicit common-success populations, and a secondary single-camera case illustrates an aggregate sparse-sampling preference with incomplete primary confirmation. The study provides a reproducible evaluation procedure for budget-dependent measurement choices, with explicit reference sensitivity and limits on generalization.

1 Introduction

A recurring measurement task is to report how far a target moved over a time window: the arc length of its trajectory, rather than the straight-line displacement between the endpoints. When the trajectory is observed only at discrete instants, the estimate is biased in two directions at once. Connecting sparse samples by straight lines cuts every corner and underestimates the arc. With non-vanishing independent position noise, dense sampling can accumulate noise-driven length, and the limiting polyline length can diverge. This conclusion is conditional on the noise model; a constant position offset does not change length.

This trade-off is established. Palmer [1] simulated it for fishing-vessel tracks and noted that the overestimation at high polling rates is inversely proportional to vessel speed. Ranacher *et al* gave a closed form for the noise-driven overestimate [2] and devoted a separate study to locating the balance point for floating-car data [3]. Noonan *et al* [4] established the full picture for animal telemetry, proved the divergence result, named the balance point the ‘Goldilocks sampling frequency’, argued that it is impractical to hit deliberately, and published a corrective estimator.

Rowcliffe *et al* [5] and Schoombie *et al* [6] quantified the two limbs empirically. We claim none of this as new and we build on it directly.

It is also known that adding cameras to a markerless motion-capture rig helps less than one might expect. Mulla *et al* [7] found that two cameras can match four for markerless hand tracking, and Walker *et al* [8] found that eighteen, twelve and six cameras change biomechanical kinematics by only a few degrees. We report the existence of saturation as confirmation of their finding, not as a discovery.

What neither literature contains is the allocation question that these two facts jointly create. A telemetry collar has one sensor and a fixed duty cycle, so the sampling interval is its only lever, and the practical response to the bias trade-off is post-hoc correction. A camera rig has two levers, and they compete for the same budget. If the cost that matters is the number of camera frames that must be transferred, decoded, detected and triangulated, then a budget of B frame reads per window satisfies

$$B = k m, \quad (1)$$

where k is the number of simultaneous views used at each instant and m is the number of instants. The design question is where to spend. The saturation studies [7, 8] vary the camera count at a fixed frame rate, and so measure only one edge of that surface.

This paper contributes the following.

1. *An explicit frame-budget comparison.* The camera count and sampling density are varied jointly. At an equal budget of 84 frames, the calibration-selected triple at 28 instants has a confirmation-set error of 9.60 mm against 50.77 mm for seven views at 12 instants. This comparison is an exploratory analysis of configurations recorded under the frozen protocol. A broader budget-cap grid includes two through seven views and exposes budget-dependent rank reversals and rare large errors.
2. *Controls on a possible explanation.* We compare an independent-pixel-noise geometry model with observed reconstruction errors, disjoint camera groups, a shared-camera noise simulation and a robust triangulation baseline. The results distinguish evidence for correlated errors from a causal or quantitative decomposition of their sources.
3. *Tests of further tuning and its costs.* We quantify the performance of one curvature-based placement rule, including an oracle-assisted version, without claiming a general optimum. We also separate typical errors from rare extreme errors and report the completion loss of robustness gates.

We follow the vision-based measurement framing of Shirmohammadi and Ferrero [9]: the camera rig is the instrument, and the object of study is the measurand and its uncertainty, not the performance of an algorithm. Section 5.6 provides a contrasting case study of a single-camera chained-registration system, where the smallest tested budget has the lowest aggregate error, with incomplete primary confirmation. This is why we present allocation as a quantity to evaluate per instrument.

2 Related work

2.1 Sampling-density bias in distance travelled

The two-sided bias, the interior optimum, and its dependence on the ratio of motion per interval to measurement noise are established in [1–4]. Noonan *et al* [4] give a practical sampling window bounded below by the ratio of error variance to velocity variance and above by the velocity autocorrelation time, prove that no estimator recovers distance when the sampling interval exceeds three times that autocorrelation time, and provide a continuous-time speed and distance estimator. Codling and Hill [10] give analytic expressions for how apparent step length scales with sampling rate in a correlated random walk; Jerde and Visscher [11] and Hurford [12] document the noise-driven corruption of step lengths and turn angles. The fractal and tortuosity literature [13, 14] is the conceptual antecedent. All of this work treats a single sensor with a fixed sampling rate.

2.2 Camera count in multi-view measurement

Triangulation uncertainty as a function of camera count, baseline and noise is characterised in [15], and diminishing returns are expected. Markerless motion-capture studies report saturation empirically [7, 8, 16]. Leizea *et al* [17] build an uncertainty budget for a multi-camera system and test the premise that more viewpoints lower uncertainty. These analyses hold the temporal sampling fixed and vary only the view count, so they do not address a shared budget.

2.3 Network design and negative results in measurement

Shen *et al* [18] derive network-configuration guidance from an error characterisation of an industrial photogrammetric system, which is the closest methodological precedent to what we do, but for a static network and a static measurand. Nasiri *et al* [19] show that a textbook optimal triangulation method is not optimal, and Das *et al* [20] show that a widely used agreement statistic misrepresents markerless versus marker-based comparison; our sections 5.3 and 5.5 are of that kind.

2.4 Positioning

Our contribution is the budget axis of equation (1), empirical allocation comparisons, and controls on possible explanations and tuning. The bias mechanism belongs to [1–4]; the existence of saturation belongs to [7, 8].

3 Measurand, measurement chain and cost model

3.1 Measurand

The arc length of the target’s trajectory between two specified instants, in metres. We report it separately from the endpoint displacement, which is a different quantity and behaves differently under sampling.

3.2 Chain

At each selected instant the target’s image position in each selected camera is triangulated by the direct linear transform into a three-dimensional point. The sequence of positions is converted to a length by one of three estimators:

- *Polygon*: the sum of the segment lengths.
- *Limited arc*: a segment-wise Hermite interpolation with a limiter that removes backward longitudinal overshoot while retaining transverse curvature, integrated by 32-point Gauss–Legendre quadrature.
- *State mean*: the length of the conditional mean path of a Wiener-velocity state-space model fitted to the selected positions, with the acceleration spectral density chosen by innovation likelihood [21]. This is the strongest of the three at realistic budgets and is the primary estimator throughout. Unless stated otherwise, the isotropic observation standard deviation is 3 mm. The acceleration spectral density is selected from 13 log-spaced values between 10^{-6} and $10^3 \text{ m}^2 \text{ s}^{-3}$ by innovation likelihood, and speed is integrated with eight quadrature points per interval. These settings are held fixed across view counts.

A fourth estimator integrates the conditional expected speed. By Fubini’s theorem the integral of the expected speed equals the expectation of the integrated speed, so this implements the conditional-expectation principle of [4] under our Wiener-velocity model. It is not a reproduction of their fitted stochastic model or model-selection procedure; section 5.4 evaluates this model-specific baseline with calibrated observation noise.

3.3 Cost model

One camera frame read at one instant is one unit, so a window costs $B = km$. Frames in which the detector returns nothing are still charged. It is a proxy for processing load, not a measured runtime or energy cost: transfer, decoding, detection and triangulation need not have equal or linear costs across implementations. Calibration is treated as a fixed setup cost. The budget deliberately does not count acquisition: a rig that records seven cameras and processes three has already paid for the sensors. The model therefore answers how much of a recorded multi-camera stream must be processed, not how many cameras to buy.

3.4 Error dispersion and reference sensitivity

For the calibration-selected triple at $m = 33$, 60 confirmation windows have a mean reference length of 6.625 m. Table 1 reports a conditional, top-down error summary. The signed bias is -8.23 mm and the standard deviation of signed errors is 7.43 mm . No bias correction is applied to any reported estimate. A correction of $+8.23 \text{ mm}$ estimated on these same windows would centre their errors by construction; its transfer to new records and its estimation uncertainty have not been validated.

The reference sensitivity proxy is the mean range of reference polyline lengths at strides one, two and four, divided by $\sqrt{3}$. Treating this range as a rectangular half-width is only a sensitivity

Table 1. Conditional error summary for the selected triple at 33 instants (99 frames per window). Dispersion and reference sensitivity are reported separately; no combined uncertainty or coverage interval is asserted.

Quantity	Evaluation	Value (mm)
Signed bias, b	uncorrected	-8.23
Error dispersion, s_e	window standard deviation	7.43
Reference sensitivity scale, s_r	decimation proxy	0.55

convention; it does not characterize optical-reference calibration, scale or synchronization uncertainty. Reference errors may already contribute to s_e , so these quantities are not combined in quadrature. Section 5.1 instead evaluates the principal comparison directly under reference decimation.

The 60 windows share four records, one rig and one calibration session. Consequently, 59 independent degrees of freedom and 95% coverage cannot be asserted. Common calibration, reference-system and timing errors may not be represented by between-window variation. A complete instrument uncertainty assessment would require those contributions, their covariance, independent calibration information and coverage validation on additional sessions. Section 5.1 instead reports paired record-level sensitivity intervals for comparisons, and section 5.2 examines temporal error structure without treating it as a full uncertainty propagation model.

3.5 Failure policy

An instant at which fewer than two views yield a detection is dropped from the polyline. If either endpoint instant is lost, the window is a failure. Failures remain in completion-rate denominators. Error summaries use completed windows; paired comparisons use the intersection of completed windows and report its size. A failed window has no finite error assigned. The earlier non-uniform sampling experiment used stricter, sequential eligibility screens; its distinct policy and conditional sample sizes are specified in section 5.3.

4 Data and frozen protocols

4.1 Primary rig

Seven hardware-synchronised cameras at 50 Hz observe a single passive marker carried through free-space motion, and an optical tracking system records the marker at 200 Hz as an independent reference [22]. We use the publisher's released per-camera two-dimensional centroids and intrinsics, with extrinsics fitted once on the separate calibration record 3. Each record gives 15 non-overlapping windows of 3.98 s, that is 200 camera frames. Development used records 19 to 23, giving 75 windows. Confirmation used records 15 to 18, giving 60 windows, previously touched only by a per-frame triangulation quality check and never by any path, sampling or estimator design.

4.2 Secondary chain, opposite regime

Three sequences from a public colour-and-depth benchmark [23] with a single moving camera and an optical tracking reference for the camera pose. Positions come from chained pairwise rigid registrations, not independent triangulation. Development used the 15 windows at offsets 1 to 17 s used by earlier work; confirmation used the 14 valid windows at offsets of at least 21 s, never previously touched.

4.3 Protocol freezing and analysis status

For each confirmation run the protocol, candidate list, camera-subset rule, decision gates and source-file hashes were frozen before the path-length confirmation predictions were computed. Predictions were sealed by a cryptographic hash before the reference was loaded for that run's scoring. The primary rig's confirmation records had already been used for per-frame triangulation quality checks, as noted above; they were not wholly unseen data. The released local timestamps and hashes document this execution sequence, rather than an independently timestamped public preregistration.

On the primary rig, the frozen candidates compared a 57-frame selected triple and a 56-frame pair against a 56-frame seven-view baseline; the one-frame mismatch is retained explicitly. Same-density comparisons and the 84-frame comparison use the recorded secondary grid. The record-resampling intervals, disjoint-group and noise controls, and robust-triangulation comparison reported here are exploratory analyses made after confirmation outcomes were examined. No equivalence margin was specified in advance. The additional budget-cap grid uses $B \in \{56, 84, 112, 168\}$ and $k = 2, \dots, 7$, with uniform $m = \lfloor B/k \rfloor$ instants. The existing

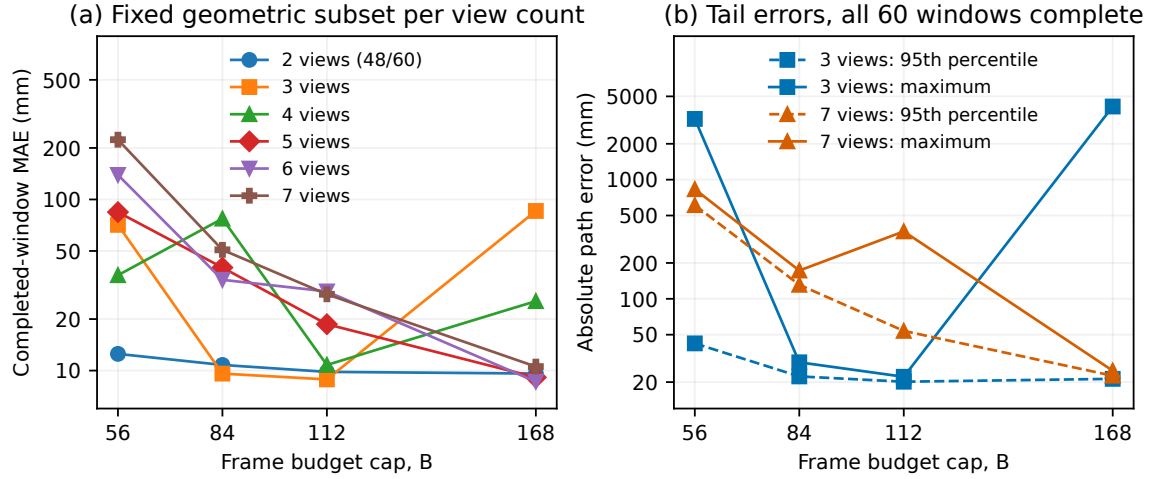


Figure 1. Exploratory budget-cap grid using one calibration-selected subset per view count and $m = \lfloor B/k \rfloor$ uniform instants. (a) Completed-window MAE; pairs complete 48/60 windows and the other view counts complete 60/60. (b) The 95th percentile and maximum error for three and seven views. These are tested schedules using nearly all the cap, not a lower error envelope over all schedules costing at most B .

Table 2. Original frozen candidate comparisons against the seven-view, eight-instant baseline (56 frames, 223.42 mm MAE, 60/60 complete). Each candidate had to reduce MAE by at least 50%, improve every record and incur no additional failures. The protocol explicitly allowed 56/57 frames. MAE is conditional on each arm's completed windows.

Views	Instants	Frames	MAE (mm)	Complete	Decision
2	28	56	12.52	48/60	Fail: completion
3	19	57	14.86	60/60	Pass

calibration-only subset score is extended to four, five and six views before computing this grid, without optimizing subsets on reference errors. Noise sensitivity at $B = 84$ uses fixed $\sigma = 0.4, 1, 3$ mm, spanning the earlier development-derived scale and the original setting. These choices were made after inspecting earlier results and remain exploratory; they are neither a new holdout nor a fresh preregistration. Of the recalculated estimates, 288 overlap the archived primary-rig estimates and reproduce them exactly. For the secondary chain, XFeat was the pre-specified primary front end and ORB was secondary; both results are reported in section 5.6.

4.4 Reference caveat

The reference length is the polyline of the 200 Hz trace between the window endpoints, itself a sampled estimate. Decimating it by two changes the reference length by a few millimetres, the scale of the smallest differences we report.

5 Results

5.1 Trading view count for sampling density

Figure 1 shows the new budget-cap grid. Before describing it, we distinguish original frozen decisions and same-density summaries. Figure 2 and table 3 summarize all subsets at fixed density on the protocol-frozen confirmation set.

Table 2 reports the original decisions, not the new exploratory grid. The triple passed all stated gates but spent one extra frame; this is not an exactly equal-cost result. The pair's completed MAE is not a like-for-like comparison to all 60 baseline windows: the baseline MAE on its 48 common windows is 270.88 mm. Its completion gate nevertheless failed. The separate 66-frame pair was not a primary candidate.

The subset medians in table 3 are close for three and seven views at $m \geq 19$, but they do not identify a deployable subset and the 35 triples are not independent replicates. We therefore report the single triple selected by the frozen calibration-only rule separately. Its camera IDs are 216f21c1, 3e0f8f0 and 44c4b2e; the rule minimizes the worst-pair geometric error scale at a calibration-derived scene centre.

At $m = 33$, record-specific differences are +0.67, +1.01, -0.02 and +0.82 mm for records 15–18. Thus the selected triple is slightly worse in aggregate, despite using 99 instead of 231 frames (43%). A nonsignificant difference at some other settings would not establish equal accuracy either. At

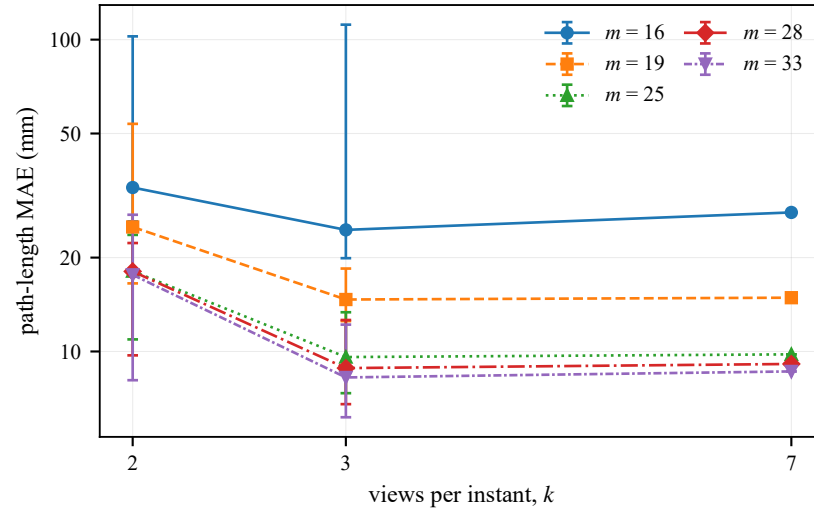


Figure 2. Error against the number of views at fixed sampling density on the protocol-frozen confirmation set. The bars span the best to the worst camera subset of that size; they are not confidence intervals. Subset medians are descriptive summaries, not the performance of the pre-selected triple.

Table 3. Confirmation set, 60 windows, state-mean estimator. Median over camera subsets of the mean absolute error, in millimetres, and the number of failed window-arms.

m	$k = 2$	$k = 3$	$k = 7$	failed, $k = 2/3/7$
16	33.6	24.5	27.9	250 / 0 / 0
19	25.1	14.7	14.9	250 / 0 / 0
25	18.1	9.6	9.8	250 / 0 / 0
28	18.1	8.8	9.1	250 / 0 / 0
33	17.6	8.3	8.6	250 / 0 / 0

Table 4. Exploratory comparison of the pre-selected triple with all seven views on 60 completed confirmation windows. Differences are three-view minus seven-view MAE. Intervals resample the four whole records with replacement (all 4^4 combinations), retaining their 15 windows each. Four clusters give limited inferential resolution; these are not equivalence tests.

m	Triple MAE	Seven-view MAE	Difference	95% resampling interval (mm)
16	72.79	27.92	+44.87	[−0.27, 134.93]
19	14.86	14.87	−0.01	[−0.41, 0.29]
25	10.08	9.79	+0.29	[−0.37, 0.70]
28	9.60	9.12	+0.48	[0.10, 0.82]
33	9.25	8.63	+0.62	[0.19, 0.93]

Table 5. Exploratory budget-cap grid: MAE in millimetres, with actual processed frames in parentheses. Each view count uses one calibration-only subset. Pairs complete 48/60 windows in every row; all other configurations complete 60/60. Full per-record, percentile and maximum errors are supplied in the archive.

Cap B	2 views	3 views	4 views	5 views	6 views	7 views
56	12.52 (56)	71.07 (54)	36.08 (56)	84.35 (55)	139.13 (54)	223.42 (56)
84	10.79 (84)	9.60 (84)	77.24 (84)	40.00 (80)	33.92 (84)	50.77 (84)
112	9.83 (112)	8.88 (111)	10.78 (112)	18.66 (110)	29.01 (108)	27.92 (112)
168	9.61 (168)	85.66 (168)	25.45 (168)	9.13 (165)	8.61 (168)	10.55 (168)

$m = 16$, one dropout leaves the selected triple with two views and produces a 3.1 m error, giving a mean of 72.79 mm rather than the 24.5 mm median over all triples. Completion alone therefore does not establish reliability. Across the 21 pairs, 250 of 1260 window-arms fail at each listed density, against zero of 2100 for the triples.

At an exactly equal cost of 84 frames, the selected triple at 28 instants gives 9.60 mm versus 50.77 mm for seven views at 12 instants, with all 60 windows completed. Every record favours the triple, with MAE differences between -46.21 and -36.03 mm; the exploratory record-resampling interval is $[-45.08, -37.25]$ mm. This supports allocating more instants on this rig within the tested grid, not a universal optimal view count. Separately, three views at 19 instants cost 57 frames and give 14.86 mm, versus 27.92 mm at 112 frames for seven views at 16 instants: a lower-cost, lower-error comparison, not an equal-cost comparison.

5.1.1 Budget-cap and parameter sensitivity. Table 5 broadens the analysis to two through seven views. All configurations use the same 60 windows and original 3 mm noise setting. Pairs complete only 48 windows, so their MAE cannot alone rank them above complete configurations. Within configurations completing all windows, the lowest observed MAE is at four views for $B = 56$, three for $B = 84$ and 112, and six for $B = 168$. These are post-hoc rankings, not validated selection rules.

At 168 frames the triple encounters two errors above 100 mm, with a maximum of 4110.45 mm, whereas the seven-view maximum is 24.92 mm. At the 56-frame cap, one triple error reaches 3233.05 mm. Completion therefore does not guarantee reliability, and using more of an available budget need not help. An upper budget permits retaining a cheaper schedule: the 84-frame triple is still feasible at a cap of 168. The table evaluates specified near-cap schedules, not the optimum over every allocation with $km \leq B$.

At 84 frames, the three/seven-view MAEs are 8.36/49.97 mm for $\sigma = 0.4$ mm, 8.48/50.04 mm for 1 mm and 9.60/50.77 mm for 3 mm, with 60/60 completions throughout. Thus this particular ordering is stable over the tested settings. These are shared isotropic-noise sensitivity settings, not individually optimized covariance models or a test of every weighted reconstruction.

For the original 3 mm setting at 84 frames, the four record differences are all negative; deleting one whole record gives differences from -42.88 to -39.49 mm. At 112 frames, the deletion range is $[-21.67, -12.62]$ mm, while at 168 frames it is $[10.14, 101.23]$ mm: the adverse ranking remains after deleting any one record. These descriptive checks do not turn four records into a larger independent sample. Resampling intervals above retain their exploratory interpretation.

Keeping each reference window's endpoints and decimating only its interior 200 Hz samples at strides one, two and four changes the 84-frame MAE difference from -41.17 to -41.14 and -41.07 mm. Its ordering also persists at the two alternative noise settings. This directly tests reference-resolution sensitivity of the main contrast; it does not test reference calibration, clock offset or absolute scale errors.

5.2 Error structure and controls on the saturation hypothesis

Saturation at a handful of cameras has been reported before [7, 8]. Table 6 summarizes position-error diagnostics on the five development records; the controls below examine possible explanations without identifying a unique mechanism.

5.2.1 The geometric prediction. Under independent isotropic pixel noise the triangulation covariance is

$$\text{Cov}(\mathbf{X}) = \sigma_{\text{px}}^2 \left(\sum_{i=1}^k \mathbf{J}_i^\top \mathbf{J}_i \right)^{-1}, \quad (2)$$

where $\mathbf{J}_i$ is the 2×3 projection Jacobian of camera i at the point $\mathbf{X}$. Evaluated along the reference trajectory, this model predicts a seven-view to three-view error ratio of 0.583 to 0.593 across the five records, that is, seven views should be about 41% more accurate.

Table 6. Predicted and realised effect of using seven views instead of three, on the five development records. The independent-pixel-noise model predicts a ratio near 0.59; the realised ratio averages 1.08. The median cosine is taken between the three-view and seven-view position error vectors.

Record	3 views RMS (mm)	7 views RMS (mm)	Realised ratio	Model ratio	Median cosine	Best six-view subset
19	3.078	2.369	0.770	0.589	0.881	drop cam. 1: 2.274 (−4.0%)
20	3.146	2.412	0.767	0.586	0.875	drop cam. 1: 2.323 (−3.7%)
21	2.992	4.415	1.476	0.593	0.869	drop cam. 6: 2.437 (−44.8%)
22	17.047	7.765	0.456	0.583	0.866	drop cam. 2: 6.411 (−17.4%)
23	2.958	5.678	1.920	0.591	0.868	drop cam. 2: 3.179 (−44.0%)

5.2.2 The realised ratio. The realised ratio averages 1.08, and on two records the seven-view solution is worse than the three-view one.

5.2.3 Correlated errors and overlapping subsets. The median cosine between three-view and seven-view error vectors is 0.866 to 0.881. This measures directional agreement, not the fraction of shared error variance. The reconstructions reuse three cameras and the same reference, so correlation alone cannot identify calibration or centroid bias.

An exploratory linearized least-squares simulation uses the projection Jacobians along every fortieth development reference point, 100 independent unit-variance Gaussian pixel-noise draws per point, and shared noise draws for cameras present in both solutions. Even with an exact reference and independent noise across cameras, the nested three/seven-view median cosine is 0.738–0.753 across records. The observed nested cosine therefore cannot be interpreted without this overlap control.

The selected three cameras and the remaining four form disjoint groups. Their observed error cosine is 0.418–0.459, compared with 0.004–0.008 in the independent-noise simulation. This is consistent with shared error sources, but common reference error and model mismatch remain alternatives; we do not identify a causal contribution or a shared-variance percentage.

5.2.4 Temporal structure and path-length sensitivity. For a polygon with segment unit directions $\mathbf{u}_i$ and position errors $\mathbf{e}_i$, the first-order length perturbation is

$$\delta L \approx \sum_i \mathbf{u}_i^T (\mathbf{e}_{i+1} - \mathbf{e}_i). \quad (3)$$

A constant translation cancels, whereas a shared but time-varying error need not. The ratio of adjacent error-increment RMS to position-error RMS for the selected triple is 0.31, 0.33, 0.37, 1.39 and 0.32 on records 19–23, respectively. Four records show temporal persistence, while record 22 contains strong increments. These position-level diagnostics do not propagate uncertainty through the nonlinear state-mean estimator or prove why its length errors saturate.

5.2.5 Sensitivity to reconstruction robustness. Removing the best single camera retrospectively reduces seven-view position RMS on each development record by 3.7% to 44.8%. This oracle removal diagnostic does not define a prospective selection rule. We additionally apply an existing pixel-consensus triangulation rule at seven views and 33 instants, using a fixed 4-pixel threshold and majority support of at least four views, without temporal gating. On the same 75 development windows, state-mean path MAE is 8.697 mm with consensus, 8.699 mm with seven-view DLT and 8.809 mm with the selected three-view DLT. All 75 windows complete; one of 2475 queried instants is rejected by consensus. Thus this control changes path error little at this density. It does not establish that all robust or weighted reconstructions behave similarly, or that additional views are intrinsically harmful. The measured allocation surface remains conditional on the calibration, estimator and detection regime.

5.3 A curvature-based non-uniform sampling rule

For a smooth curve the deficit of one segment is approximately

$$\Delta_i \approx \frac{(\Delta t_i)^3}{24} \kappa_i^2 \|\mathbf{v}_i\|^3, \quad (4)$$

with κ the curvature and $\mathbf{v}$ the velocity. Write $a(t) = \kappa(t)^2 \|\mathbf{v}(t)\|^3$ and let $n(t)$ be the number of intervals per unit time. In the small-interval approximation, the total polygon deficit is

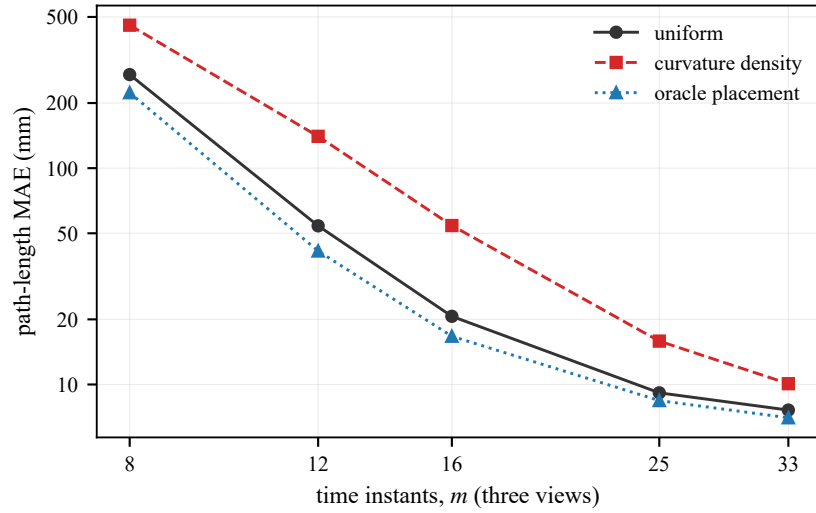


Figure 3. Non-uniform placement of instants on the development set, at three views. The reference-assisted rule improves completed-window MAE by approximately 7.9% to 23.5% in the plotted three-view cells. The deployable rule is worse on their respective common-success populations.

Table 7. Non-uniform placement on a pool of 75 development windows. State-mean MAEs (mm) use only the common-success windows in the last-column denominator. The numerator counts windows where the deployable rule beats uniform placement. Eligibility and successful-output counts are given below.

k	m	Uniform	Curvature	Oracle	Deployable wins
3	8	270.7	458.0	223.2	12 / 67
3	16	20.7	54.3	16.7	7 / 63
3	25	9.2	15.9	8.4	10 / 61
3	33	7.6	10.1	7.0	18 / 62
7	8	242.9	425.5	200.9	13 / 75
7	33	7.4	9.8	6.6	20 / 75

$\int_0^T a(t)/(24n(t)^2) dt$, subject to $\int_0^T n(t) dt = m - 1$. Variation with a Lagrange multiplier gives $-a/(12n^3) + \lambda = 0$, hence $n(t) \propto a(t)^{1/3}$. This continuous density problem, rather than minimization of a fixed-coefficient discrete sum over interval lengths, gives the rule

$$\rho(t) = (\kappa^2 \|\mathbf{v}\|^3)^{1/3} = \frac{\|\mathbf{v} \times \mathbf{a}\|^{2/3}}{\|\mathbf{v}\|}. \quad (5)$$

Unlike a telemetry collar, a camera system can actually execute such a placement, so this is the natural next lever after the budget split.

We tested three arms at identical cost: uniform placement; a deployable arm that spends half the budget on a uniform probe, fits the state model, computes ρ from the smoothed trajectory and places the remaining instants at equal increments of the cumulative ρ while keeping the probe instants; and an oracle arm that computes ρ from the reference trajectory. The oracle uses otherwise unavailable trajectory information to evaluate this particular rule. It is neither deployable nor a certified optimum for path-error minimization, even within curvature-driven methods. Figure 3 and table 7 give the result.

The recorded successful outputs for uniform/deployable/reference-assisted arms, respectively, are 71/69/67, 66/64/64, 67/62/63 and 66/63/62 out of 75 for the listed three-view densities; all three arms have 75 outputs in the listed seven-view cells. The common sets contain 67, 63, 61 and 62 windows. Unlike the main experiment, this earlier implementation requires every selected instant to be valid and screens uniform, probe and later stages sequentially. Some absent downstream outputs were never attempted, so these counts are not independent per-method completion probabilities. The comparisons are conditional on joint eligibility, not a claim about overall operational success. The additional noise-calibration burst is a setup resource excluded from the within-window frame count.

The deployable rule has higher aggregate MAE in every tested cell, by 30% to 170%. Two possible explanations, not isolated by ablation, are that equation (4) holds only for small Δt , so concentrating instants in high-curvature regions creates long gaps whose cubic penalty may outweigh local gains; and curvature estimated from a half-budget probe may be noisy.

Table 8. Development set at three views, state-mean estimator with calibrated noise, in millimetres, as a function of the number of instants. The median broadly falls then stabilizes, increasing slightly from 5.8 to 6.1 mm at the final setting; the ninetieth percentile falls. The mean rises mainly because of one window at 50 instants and one at 200.

Statistic	8	16	25	33	50	100	200
Mean	269.6	21.1	10.0	7.9	29.4	6.5	35.2
Median	114.0	16.8	8.8	6.6	6.1	5.8	6.1
90th percentile	592.8	41.7	20.6	15.7	13.5	12.6	11.5
Maximum	1893.7	60.6	29.2	26.8	1691.6	20.3	2182.5
Windows over 100 mm	38	0	0	0	1	0	1

Reference-assisted curvature placement improves on uniform sampling by 8% to 25%, with a median of 17% across the tested cells. This describes one rule and one grid. The small-interval polygon approximation does not optimize the final state-mean estimator’s absolute error. An earlier reference-based enumeration gave about 35% improvement by selecting instants against the final error directly. The gap cannot be assigned quantitatively to error cancellation or to recoverable signal from these experiments. Neither experiment bounds what another adaptive or learned rule could gain.

5.4 A calibrated continuous-time estimator does not remove the effect

A natural objection is that the corrective estimator of [4] should be used instead of characterising the allocation. We evaluate its conditional-expected-speed principle within our state model, rather than reproduce the full method. We calibrated the observation noise from a dense consecutive burst at the native frame interval, assuming that measurement noise dominates process noise, obtaining 0.20 to 0.43 mm per record and subset.

Calibrating the noise gives a small consistent improvement to the state-mean estimator, from 8.81 to 7.86 mm at three views and 33 instants. The conditional-speed estimator does reduce sensitivity to sampling density at the sparse end, as designed: at seven views and eight instants it gives 165.2 mm against 242.9 mm. But it carries a large positive bias at every setting tested, +42 to +180 mm, and is dominated in absolute error by the conditional-mean-path estimator throughout. We attribute this to model misspecification rather than to the method: [4] uses an Ornstein–Uhlenbeck process with a fitted velocity autocorrelation time and selects the model by a corrected information criterion, whereas our Wiener-velocity model has none. We therefore do not claim that the method of [4] fails, only that with our state model it does not remove the allocation effect in this implementation. This comparison does not rule out a better-specified stochastic model. The dense probe used for noise calibration is an additional calibration resource, outside the per-window frame budget.

5.5 Rare extreme errors dominate the dense-regime mean

The mean absolute error over our windows rises at high sampling density, which invites the interpretation that the classical noise-inflation limb has been reached. The distribution, in table 8 and figure 4(a), shows the importance of rare extreme errors.

The rise in the mean at 50 and at 200 instants is produced by a single window each, in which one instant received a grossly wrong triangulation. This is outlier exposure: denser sampling draws more instants and has more opportunities to include a bad one. An order-of-magnitude estimate of the classical mechanism for this rig, with calibrated white noise near 0.4 mm and about 34 mm of motion per 20 ms step, gives roughly 2 mm of inflation at the densest setting, consistent with the median flattening near 5 to 6 mm rather than rising.

These observations motivate robustness checks without excluding simultaneous noise accumulation. A combined view-consensus and Kalman-innovation gate reduces the median completed-window MAE across 35 triples from 38 mm to 6 mm at 200 instants and improves 34 triples. It rejects inconsistent observations, and rejects a window if an endpoint is lost. Only 2432 of 2625 window-arms complete (92.6%), compared with 2625 without the gate; 193 failures accompany the error reduction. The error summaries therefore condition on different completed populations and do not isolate the effect of temporal gating. The same gate hurts some lower-density settings by rejecting real motion. A deployment decision must assess completion and error together. Mean error alone cannot distinguish smooth noise accumulation from rare gross failures.

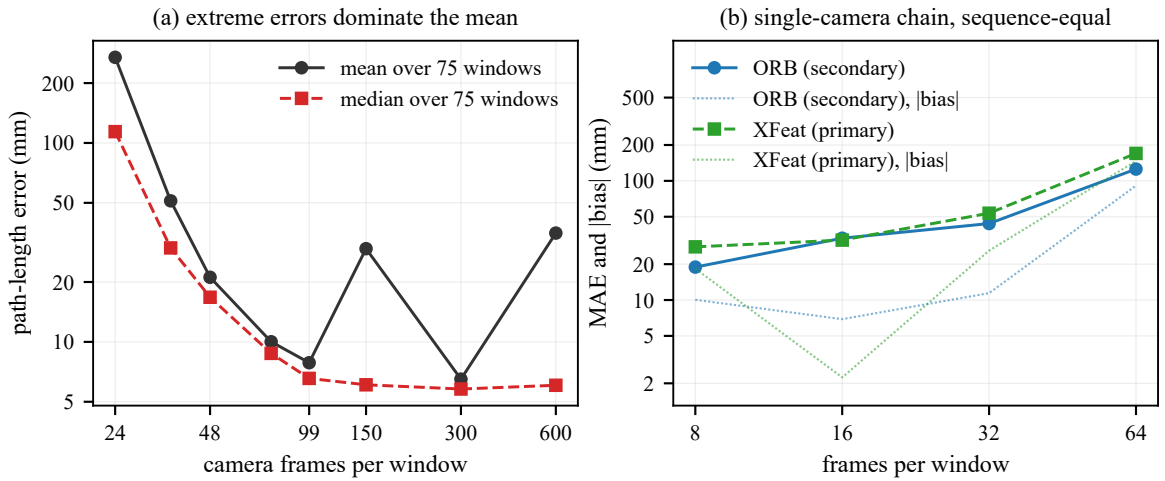


Figure 4. (a) Rare extreme errors dominate the dense-regime rise in the mean; the median remains near 6 mm. (b) The tested single-camera chain favours fewer frames in aggregate, and the magnitude of the signed bias reaches a minimum between 16 and 32 frames, where the bias changes sign.

Table 9. Secondary chain, 14 confirmation windows. MAE in millimetres with signed bias in parentheses; both are averaged within each sequence and then equally across the three sequences (2, 7 and 5 windows). XFeat is primary; ORB is secondary.

Front end	8 frames	16	32	64
ORB (secondary)	18.9 (−10.0)	33.0 (−6.9)	43.8 (+11.4)	125.6 (+91.1)
XFeat (primary)	27.9 (−18.3)	31.8 (−2.2)	53.5 (+26.0)	170.0 (+146.8)

5.6 The opposite regime: chained single-camera egomotion

In the secondary chain the positions come from chained pairwise rigid registrations of a single moving camera. Each edge contributes registration noise that does not shrink with more edges, while the roughly 0.8 m of smooth motion in a 4 s window is nearly fully captured by eight instants. Figure 4(b) and table 9 give the result on 14 unseen windows.

The frozen primary criterion required XFeat’s sequence-equal MAE at eight frames to beat both 32 and 64 frames overall, not exceed the 32-frame error in any sequence, and complete at least as many windows as at 32 frames. It was not fully met: the aggregate and completion conditions passed, but the per-sequence condition failed. ORB passed the same criteria as a pre-specified secondary analysis. Both front ends completed all 14 windows at every budget. Thus the aggregate trends and the secondary result support a sparse-sampling preference in this chain, but the primary confirmation does not establish it across all sequences. The signed bias changes sign between 16 and 32 frames, consistent with competing sampling and registration effects. Fourteen windows from three sequences on the same rig do not establish transfer to other instruments.

6 Limitations

- *Two instruments, each a single rig.* The budget-cap grid has different best observed view counts and rank reversals. No universal three-view optimum or saturation threshold is established. The controls in section 5.2 do not identify a unique causal mechanism; calibration, reference error, estimator choice and target visibility can all affect the observed surface.
- *Published detections, not raw pixels.* For the multi-camera rig we use the publisher’s released centroids, so detection cost and detection failure modes are inherited rather than measured.
- *The cost model counts processing, not acquisition.* A capture-design reading, namely to buy three cameras rather than seven, is not supported, because a rig built with three cameras would have different occlusion behaviour.
- *Reference resolution.* Differences of one to three millimetres are within the sensitivity of the reference polyline to its own sampling.
- *One target, free-space motion.* No occlusion-heavy, multi-target or articulated cases are covered.

- *One sampling rule.* The reference-assisted rule in section 5.3 is not an upper bound on other curvature-based, adaptive or learned schedules.
- *Limited confirmation.* The record-level analyses have only four clusters, and the primary single-camera criterion was not fully met. The additional sensitivity analyses are exploratory, not fresh confirmations.

7 Conclusion

For this synchronised rig, allocating 84 frames to three views at 28 instants gives lower observed MAE than seven views at 12 instants, and this contrast persists under the tested noise settings, reference decimation and whole-record deletion. The broader two-to-seven-view budget grid nevertheless reverses the three/seven-view ranking at 168 frames because of rare large errors. Thus a budget increase is not a prescription to sample more densely, and the lowest-error tested schedule is not a universal camera-count rule. At a fixed 33 instants the selected triple uses 43% of the frames but has slightly larger error; equivalence remains unestablished.

The reusable procedure is to specify a processing cap and failure policy, select camera subsets without using evaluation errors, and compare feasible schedules using error, completion, tail risk and sensitivity together. A prospective deployment would require selection on development data followed by validation on new records or sessions; the new grid here is exploratory. Correlated-error diagnostics do not identify a unique mechanism, and the curvature-rule comparisons condition on explicit joint eligibility. The single-camera example illustrates a contrasting regime with incomplete primary confirmation. These qualifications delimit the measurement choices supported by the present evidence.

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Author contributions

Jun Ji: data curation, methodology, formal analysis, resources, writing – original draft. **Zihan Li:** software, formal analysis, writing – review and editing. **Bowen Tan:** software, formal analysis, writing – review and editing. **Shengjie Guo:** supervision, validation, writing – review and editing. **Yi Sui:** software, formal analysis, validation, writing – review and editing. **Xiaolei Zhang:** supervision, project administration, writing – review and editing. **Yi Li:** supervision, project administration, writing – review and editing.

Data availability

The two datasets analysed here are third-party public releases, cited as [22] and [23]; we redistribute neither, and the access routes are those given by their publishers. The analysis code, frozen protocols, input hash lists, sealed predictions, scoring records and revised manuscript are archived as version 1.0.3 at <https://doi.org/10.5281/zenodo.22794671>. The matching source release is <https://github.com/Jun-Jason-Ji/views-or-instants/tree/v1.0.3>. The version preserves the original protocols and sealed results, alongside the exploratory analysis update with explicit sequence-equal aggregation, record-level comparisons and exploratory geometry controls, with input hashes and a reproducible analysis script. Protocol freezing and prior data exposure are described in section 4.

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